TEMASEK JUNIOR COLLEGE

PRELIMINARY EXAMINATIONS

HIGHER 1

CANDIDATE NAME	
CIVICS GROUP	/
CENTER NUMBER	S
INDEX NUMBER	

CHEMISTRY **8873/02**

Paper 2 Structured Questions **23 August 2018**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Civics Group, centre number, index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	/ 60
1	/ 16
2	/ 13
3	/ 7
4	/ 8
5	/ 6
6	/ 10
Section B	/ 20
7	/ 20
8	/ 20
Paper 1	/ 30
Total	

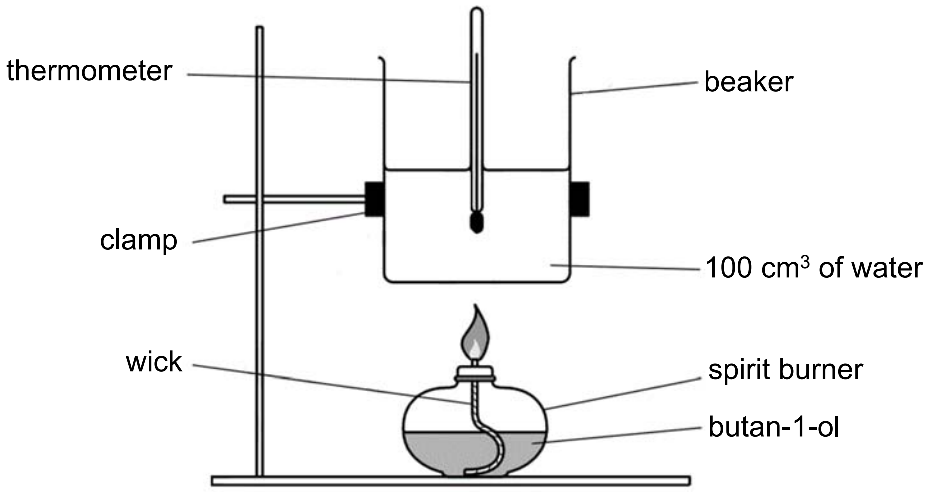
This document consists of **29** printed pages.

Section A

Answer **all** the questions in this section in the spaces provided.

1 **(a)** For many compounds, the enthalpy change of formation cannot be calculated directly. An indirect method based on enthalpy change of combustion can be used.

The enthalpy change of combustion of butan-1-ol can be found by a calorimetry experiment in which the heat given off during combustion is used to heat a known mass of water and the temperature change recorded.



temperature of water before heating = 25.0 °C

temperature of water after heating = 66.1 °C

mass of spirit burner and butan-1-ol before heating = 80.44 g

mass of spirit burner and butan-1-ol after heating = 79.70 g

(i) The heat transfer from the spirit burner to water is known to be only 70% efficient. From the data provided above, determine the enthalpy change of combustion of butan-1-ol.

[3]

(ii) Write the chemical equation that represents the standard enthalpy change of combustion of butan-1-ol, including the state symbols.

.... [1]

(iii) Use the bond energies given in the Data Booklet to calculate another value for the standard enthalpy change of combustion of butan-1-ol.

[2]

(iv) Suggest a reason for the discrepancy between this value and that calculated in **(a)(i)**.

....

....

.... [1]

(b) The table below gives some data on four fuel sources: methanol, ethanol, hydrogen and octane. Octane can serve as a rough approximation of petrol.

name	formula	molar mass/ g mol^{-1}	density/ g cm^{-3}	$\Delta H_c^\ominus$ (298K)/ kJ mol^{-1}
methanol	CH_3OH	32	0.793[a]	-726.0
ethanol	CH_3CH_2OH	46	0.789[a]	-1367.3
liquid hydrogen	H_2	2	0.0711[b]	-285.8[#]
octane	C_8H_{18}	114	0.703[a]	-5470.2

[a] At 298K and 1 bar pressure

[b] At 20K and 1 bar pressure

[#] standard enthalpy change of combustion of hydrogen **gas**

(i) Calculate the density of gaseous hydrogen at 298 K and 1 bar pressure. Assume 1 mol of any gas occupies 24.7 dm^3 at 298 K and 1 bar pressure. Give your answers in g cm^{-3}.

[1]

(ii) An important property of a fuel, especially when the fuel has to be lifted such as in aviation, is the energy released on combustion per gram of fuel.

Calculate the enthalpy change of combustion per gram of fuel at 1 bar pressure and 298K for methanol.

[1]

(iii) Another important characteristic of a fuel, especially when there is a fuel tank of limited size, is the energy released on combustion per cm^3 of fuel.

Calculate the enthalpy change of combustion per cm^3 of fuel for octane.

[1]

(iv) Explain why, given the data in the question, it is not strictly possible to make a fair comparison of the energy released per cm^3 of liquid hydrogen with other fuels.

....

....

.... [1]

(c) Carbon monoxide and steam are products of incomplete combustion of hydrocarbons. An equilibrium mixture of carbon monoxide and steam is also produced when carbon dioxide is mixed with hydrogen gas.

$$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$$

(i) 0.0100 mol of CO_2 and 0.0200 mol of H_2 were mixed and allowed to reach equilibrium at a temperature T. The final mixture contained 6.93×10^{-3} mol of H_2O.

Calculate the value of K_c at this temperature

[2]

(ii) The numerical value of equilibrium constant is 0.08 at 400 °C and 0.40 at 600 °C. Explain whether the reaction between carbon dioxide and hydrogen is exothermic or endothermic.

…………………………………………………………………………………………..

…………………………………………………………………………………………..

…………………………………………………………………………………………..

………………………………………………………………………………………… [2]

(iii) State and explain the effect of increasing pressure on the position of this equilibrium.

…………………………………………………………………………………………..

…………………………………………………………………………………………..

………………………………………………………………………………………… [1]

[Total: 16]

2 **(a)** What do you understand by the *Brønsted–Lowry* theory of acids and bases?

………………………………………………………………………………………… [1]

(b) The following is a list of compounds that react with or dissolve in water:

Sodium chloride, hydrogen bromide, silicon tetrachloride, ammonia, methanol.

Water can react as either an acid or a base. Choose a compound from the above list with which water acts as

- a *Bronsted* base
- a *Bronsted* acid

Construct a balanced equation for each reaction.

• water as a Bronsted base

• water as a Bronsted acid

[2]

(c) Lactic acid is a monoprotic acid which is an important flavouring component of many foods such as cheese, yoghurt and pickled cabbage. A solution of lactic acid in water containing 0.10 moldm^{-3} has a pH of 2.43.

(i) Is lactic acid a strong or weak acid? Explain your answer.

…………………………………………………………………………………………..

…………………………………………………………………………………………..

………………………………………………………………………………………… [1]

(ii) Use the data given to calculate the value of K_a for lactic acid.

[2]

(iii) Suggest a suitable indicator for the titration of lactic acid with aqueous barium hydroxide.

.... [1]

(iv) A sample of lactic acid was extracted from a blood sample. It was dissolved in water and titrated with 0.250 mol dm^{-3} barium hydroxide. It was found that 22.3 cm^3 of hydroxide was required for neutralisation.
Calculate the mass of lactic acid in the blood sample.
(M_r of lactic acid = 90.0)

[2]

(v) The mass composition of carbon, hydrogen and oxygen in lactic acid is 40.0%, 6.7% and 53.3% respectively. Determine the molecular formula of lactic acid.

[1]

(vi) When lactic acid is strongly oxidised, it gives an organic product with the same number of oxygen atoms. State the reagent and condition for the strong oxidation to occur.

Write a balanced equation for the oxidation reaction showing clearly the structural formula of lactic acid and its oxidation product.

Reagent and condition: ………………………………………………………………….

Equation:

[2]

(d) Explain, using an equation why an aqueous mixture of lactic acid and sodium lactate can act as a buffer solution on addition of alkali.

You may represent lactic acid as HA and sodium lactate as Na^+A^-.

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………. [1]

[Total: 13]

3 **(a)** Outline the principles of the Valence Shell Electron Pair Repulsion (VSEPR) theory.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

………………………………………………………………………………………… [2]

(b) Predict and draw the shapes of the following molecules or ions.

(i) F_2O

[1]

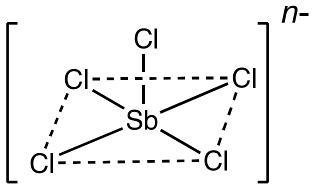
(ii) H_3O^+

[1]

(iii) ClF_4^-

[1]

(c) Antimony, Sb, is in Group 15 of the Periodic Table. It forms a series of salts which contain the $SbCl_5^{n-}$ anion, the structure of which is square pyramidal.



Deduce the total number of electrons around the antimony atom, the value of *n* and the oxidation number of Sb in this ion.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

………………………………………………………………………………………… [2]

[Total: 7]

4 **(a)** Ammonium chloride, NH_4Cl, is a salt which has covalent bonding, dative bonding and ionic bonding. It is very soluble in water and can be used to maintain the urine at an acidic pH in the treatment of some urinary-tract disorders.

(i) What is a dative bond?

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [1]

(ii) Draw a "dot-and-cross" diagram to show the bonding in NH_4Cl.

[1]

(iii) In terms of structure and bonding, explain why NH_4Cl is highly soluble in water.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [2]

(b) **(i)** Research in nanotechnology is growing and many new nanomaterials are being produced with ever more uses.

Suggest what is meant by the term nanomaterials.

………………………………………………………………………………………………

……………………………………………………………………………………………… [1]

(ii) Describe the structure of graphene. Explain how this structure gives graphene the properties of high strength and high electrical conductivity.

You may find it useful to include a labelled diagram in your answer.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

……………………………………………………………………………………………… [3]

[Total: 8]

5 Period 3 elements react with oxygen to form oxides.

(a) Write equations for the reactions, if any, of the following three oxides with water. In each case, describe the effect of the resulting solutions on Universal Indicator solution.

(i) Na_2O

…………………………………………………………………………………………………

…………………………………………………………………………………………… [1]

(ii) Al_2O_3

…………………………………………………………………………………………………

…………………………………………………………………………………………… [1]

(iii) P_4O_6

…………………………………………………………………………………………………

…………………………………………………………………………………………… [1]

(b) The oxides MgO, Al_2O_3 and SiO_2 are all used as refractory materials due to their high melting points. The last two are major constituents of gemstones, such as rubies, sapphires and amethysts.
If a sample of one of the oxides was provided as a white powder, describe the reactions you could carry out on the powder to determine which of the three oxides it was.

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

………………………………………………………………………………………………… [3]

[Total:6]

6 **(a)** Some of the most commonly used polymers are formed by the polymerisation of ethene, C_2H_4.The presence of side-chains affects the bulk properties of an addition polymer.

Poly(ethene) exists in two different forms LDPE (low density poly(ethene)) which has lots of side-chains, and HDPE (high density poly(ethene)) in which there are fewer and shorter side-chains.

(i) Explain with the aid of sketches why the presence of side-chains causes a difference in density in poly(ethene).

…………………………………………………………………………………..…………..

……………………………….……….………………………………………….……..

…………………………………..……………………………………………………………

……………………………………………….………………………………… [2]

(ii) By reference to the type of bonding between the poly(ethene) chains, explain why LDPE has a lower melting point that HDPE.

…………………………………………………………………………………..…………..

……………………………….……….………………………………………….……..

…………………………………..……………………………………………………………

……………………………………………..………………………………..…… [1]

(b) Poly(vinyl alcohol), PVA, is a synthetic polymer used in papermaking, textiles, eye drops (such as artificial tears for dry eyes) and a variety of coatings. The monomer of PVA is vinyl alcohol, CH_2CHOH.
Poly(vinyl chloride), PVC, is a synthetic water-resistant polymer used for producing water pipes, textiles and clothings such as raincoats. The monomer of PVC is vinyl chloride, CH_2CHCl.

(i) Draw a section of the PVA and PVC molecules containing at least 2 monomer molecules, and identify clearly the repeat units with [ ].

A section of the PVA:

A section of the PVC:

[2]

(ii) What type of polymerisation reaction is likely to have produced these polymers?

………………………………………………………………………………………….. [1]

(iii) With reference to the structure and bonding, explain why PVA is commonly used in eye drops while PVC is used in producing clothings such as raincoats.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [2]

(iv) Suggest a reason why PVC might break down when exposed to concentrated aqueous sodium hydroxide whereas poly(ethene) does not.

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [2]

[Total:10]

Section B

Answer **one** questions from this section, in the spaces provided

7 **(a)** Hydrogen peroxide is a mild antiseptic used on the skin to prevent infection of minor cuts, scarpes and burns.

A stock solution of H_2O_2 was diluted by adding 40.0 cm^3 of the solution into a standard flask which was filled with distilled water to make a 250 cm^3 standard solution. 25.0 cm^3 of the standard solution was titrated with 0.200 mol dm^{-3} of acidified $KMnO_4$ solution. 22.50 cm^3 of $KMnO_4$ solution was required to reach the end point and oxygen gas was produced at the end of reaction.

(i) Write a balanced equation for the reaction between hydrogen peroxide and acidified $KMnO_4$ solution, showing your working clearly

....

....

....

.... [1]

(ii) Determine the amount of H_2O_2, in moles, that reacted with $KMnO_4$.

[1]

(iii) Hence, determine the concentration of H_2O_2 in the stock solution.

[2]

(b) Hydrogen peroxide reacts with acidified iodide ions to liberate iodine according to the following reaction:

$$H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow 2H_2O(l) + I_2(aq)$$

The rate for the reaction between iodide and aqueous H_2O_2 was studied using two solutions of different initial concentrations of H_2O_2 and the same initial concentration of iodide. The results are as shown below:

Time / s	Experiment 1	Experiment 2
	$[I^-]$ / mol dm^{-3} when $[H_2O_2]$ = 0.20 mol dm^{-3}	$[I^-]$ / mol dm^{-3} when $[H_2O_2]$ = 0.30 mol dm^{-3}
0	0.0100	0.0100
40	0.0079	0.0070
80	0.0062	0.0049
120	0.0049	0.0034
160	0.0038	0.0024
200	0.0030	0.0017

(i) Plot a graph of these results, putting all the data on the same axes. Label each curve clearly.

[2]

(ii) From the graphs, deduce the order of reaction with respect to iodide and aqueous hydrogen peroxide.

[4]

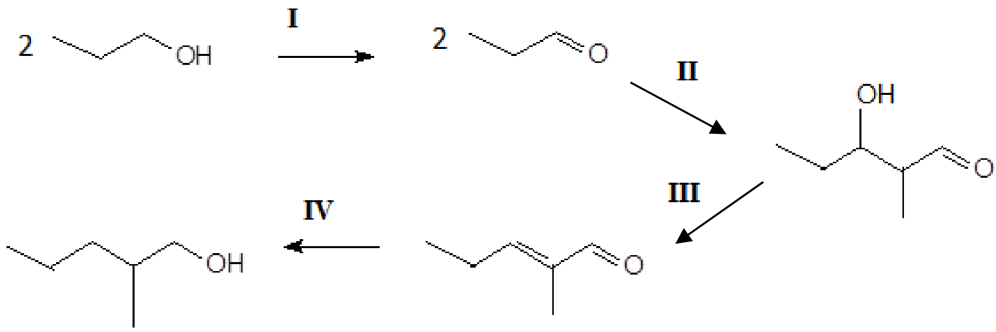
(iii) The order of reaction with respect to H^+ is zero. Hence, write the rate equation for the reaction.

………………………………………………………………………………………. [1]

(iv) Calculate the rate constant for **experiment 2**, stating its units.

[1]

(c) A four–step process to convert propanol into 2-methylpentan–1–ol could be an important step for production of renewable energy.



Compound G

(i) State the reagents and conditions required for steps **I** and **IV**.

Step I : ....

....

Step IV : ....

....

[2]

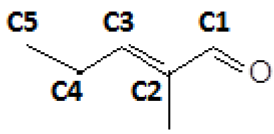
(ii) By considering the change in molecular formulae shown in steps **II** and **III**, suggest the types of reaction that occurred in both steps.

- **step II:** ....
- **Step III:** .... [2]

(iii) Draw the skeletal formula of the other stereoisomer of compound **G**.

[1]

(iv) By considering the hybridisation of the carbon atoms labelled **C1**, **C2**, **C4** and **C5** in compound **G** below, predict which sigma bond, **C1-C2** or **C4-C5,** will be shorter and explain your answer.



Compound **G**

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [2]

(v) Draw the hybrid orbitals around the carbon atom labelled **C4**.

[1]

[Total: 20]

8 **(a)** Describe the thermal decomposition of the hydrogen halides HCl, HBr and HI and explain any variation in their thermal stabilities.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [3]

(b) Halogenoalkanes can be made by the reaction of the corresponding halogen with alkanes.

When propane is treated with a small quantity of chlorine in the presence of ultraviolet light, two mono-substituted products are produced.

The order of reactivity of tertiary, secondary and primary hydrogen atoms follows a 5:4:1 ratio.

Draw the structural formula of the two mono-substituted products, and state the approximate ratio in which they are formed.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [2]

(c) Compound **A** (M_r=116) is a sweet smelling substance. Acid hydrolysis of compound **A** yields two products, compounds **B** and **C**. **C** reacts with Na_2CO_3 to give CO_2.

A series of tests were performed on compound **B**.

(i) **B** has the molecular formula $C_4H_{10}O$. Reacting **B** with acidified potassium manganate(VII) decolourises the solution and produces compound **D**, $C_4H_8O_2$. Heating **B** with excess concentrated sulfuric acid produces compound **E**, which does not show cis-trans isomerism. Reaction of **E** with hydrogen gas forms 2-methylpropane.

Suggest structures for **B, D** and **E** and explain the reactions involved.

....

....

....

....

....

....

....

....

....

Compound	Structure
B	
D	

E	

[5]

(ii) Hence, deduce a possible structure for compound **A**.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

Structure of compound A	

[2]

(d) A student wishes to investigate the kinetics of the acid-catalysed hydrolysis of compound **A** described in **(c)**.

(i) A number of experiments at a constant pressure of 101 kPa and temperature of 90 ^{o}C were performed in which the concentrations of compound **A** and hydrogen ions were varied. The results are shown below:

Experiment	**[H^+] / moldm^{-3}**	**[compound A] / moldm^{-3}**	**Initial rate/ mol dm^{-3}s^{-1}**
1	0.0100	0.120	1.8×10^{-4}
2	0.0050	0.240	1.8×10^{-4}
3	0.0050	0.480	3.6×10^{-4}

Using the above data, deduce the order of reaction with respect to compound **A** and hydrogen ions.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………… [2]

(ii) Hence, construct a rate equation and use it to calculate a value of the rate constant, k, including its units, for the acid-catalysed hydrolysis of compound **A**.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………… [2]

(iii) With reference to a diagram where necessary, state and explain how the rate of reaction would change when the experiment was conducted:

- in the absence of hydrogen ions

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

- at a pressure of 121 kPa

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

…………………………………………………………………………………… [4]

[Total: 20]